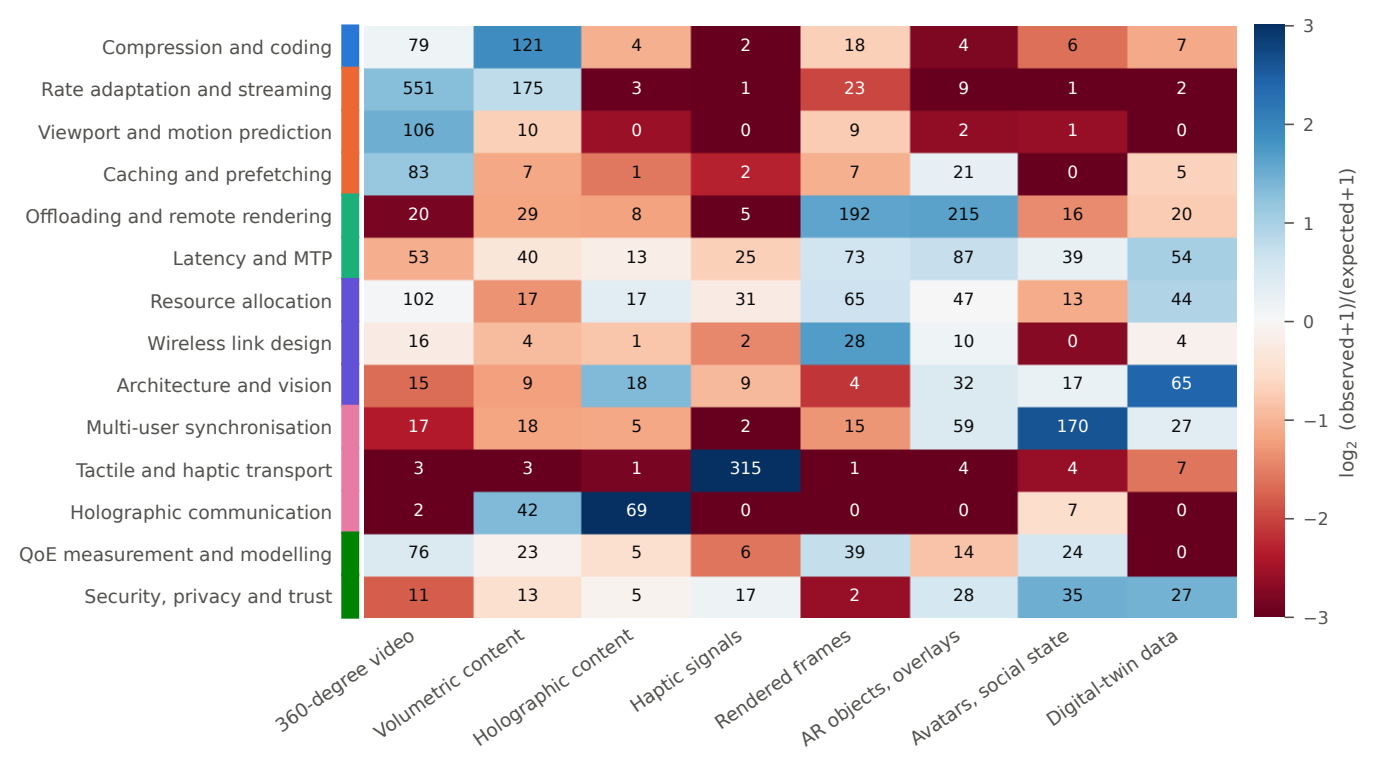
Compression and coding
Rate adaptation and streaming
Viewport and motion prediction
Caching and prefetching
Offloading and remote rendering
Latency and MTP
Resource allocation
Wireless link design
Architecture and vision
Multi-user synchronisation
Tactile and haptic transport
Holographic communication
QoE measurement and modelling
Security, privacy and trust
79 121 4 2 18 4 6 7
551 175 3 1 23 9 1 2
106 10 0 0 9 2 1 0
83 7 1 2 7 21 0 5
20 29 8 5 192 215 16 20
53 40 13 25 73 87 39 54
102 17 17 31 65 47 13 44
16 4 1 2 28 10 0 4
15 9 18 9 4 32 17 65
17 18 5 2 15 59 170 27
3 3 1 315 1 4 4 7
2 42 69 0 0 0 7 0
76 23 5 6 39 14 24 0
11 13 5 17 2 28 35 27
360-degree video
Volumetric content
Holographic content
Haptic signals
Rendered frames
AR objects, overlays
Avatars, social state
Digital-twin data
log₂ (observed+1)/(expected+1)
3
2
1
0
−1
−2
−3